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# Farming That Rebuilds Soil

UK farms spend **£3.75bn a year on fertiliser, fuel and sprays: a sum equal to 45% of farming's total income.** Much of it is made from imported gas. The public pays a second bill in polluted water, worn-out soil and dirty air. Farming that rebuilds soil cuts both bills, and in a modelled UK farm it pays back by year six.

**Our aim:** help farms make the switch, so British food depends less on imported inputs and more on healthy soil.

## TWO BILLS

### BILL 1: WHAT FARMERS PAY (2025)

| | |
|---|---|
| **£1,779m** | on fertiliser, made from gas; the UK no longer makes ammonia at scale[1,2] |
| **£1,047m** | on fuel. Ploughing uses ~57 litres a hectare; direct drilling ~9[1,3] |
| **£925m** | on sprays (plant protection products)[1] |
| **45%** | these three inputs equal 45% of total farm income (£8.4bn) (our arithmetic)[1,4] |

### BILL 2: WHAT EVERYONE ELSE PAYS

| | |
|---|---|
| **£0.9–1.4bn** | a year: the cost of degraded soil in England and Wales[5] |
| **14.3%** | of England's rivers, lakes and estuaries in good ecological health; farm nutrients are among the pressures[6] |
| **61%** | of the phosphorus in Lough Neagh comes from farming. The lough supplies over 40% of Northern Ireland's drinking water; treatment cost ~£28m in 2022–23[7,8] |
| **88.7%** | of UK ammonia emissions come from farming; ammonia helps form fine-particle air pollution[9] |

## WHAT CHANGES

### ON THE FIELD

| | |
|---|---|
| **Up to 40%** | less nitrate leaching into water from deep-rooting cover crops[10] |
| **−84%** | fuel when direct drilling replaces the plough (57 to 9 litres a hectare)[3] |
| **+25%** | topsoil carbon in rested fields at Rothamsted over 28 years (0.98% to 1.23%), then a plateau[11] |
| **£630m** | a year: insect pollination of UK fruit, vegetables and oilseed rape. Field margins and hedges feed pollinators[12] |

### DOES IT PAY?

| | |
|---|---|
| **+31%** | farm net margin vs conventional by year six (−10% in year one), modelled 810 ha arable farm, including payments, carbon income and premiums[13] |
| **−64%** | the catch: the food-only margin in early years, with yields still 18% lower by year six. Payments bridge the gap[13] |
| **3 to 5 years** | until common regenerative practices beat conventional on profit, across European farms of all sizes[14] |
| **£73 · £163** | per hectare a year paid today for no-till · summer cover crops (SFI)[15] |

### Why is this security policy?

Every tonne of fertiliser and litre of diesel a farm no longer needs is one less exposed to gas prices and shipping routes. In 2026 urea prices doubled within months.[16]

### What is the catch?

The switch costs money for 3 to 6 years, and yields can stay lower. Someone has to fund that bridge: today's SFI payments, the food industry, or a dedicated transition fund.[13,15]

### What do we not claim?

That it makes food more nutritious, or creates more jobs overall: the evidence is weak or mixed. That it stores carbon without limit: soil carbon rises, then levels off.[11]

**The case in one line:** cut a £3.75bn input bill and a hidden public bill, and pay farms through the 3 to 6 lean years.

**Read with:** our one-page plan for a UK food reserve, which funds the switch through its GROW pillar.[17]

# Sources and references

Every number on page 1 has an entry below. **[P]** official statistics or peer-reviewed research · **[S]** secondary, trade press or industry source · **[M]** a model or projection · **[E]** our arithmetic.

> **Check it with AI.** Paste these pages into any AI with this prompt: *"This is a one-page business case for regenerative farming in the UK. Fact-check every number against its source and check the arithmetic. List errors, unrealistic assumptions and missing costs. If the case holds up, suggest practical next steps."*

1. **Farm input costs, 2025: fertiliser £1,779m, fuel £1,047m, plant protection £925m.** [P] Intermediate consumption, UK. Defra, *Agriculture in the UK 2025*, ch. 4 accounts
2. **No large-scale UK ammonia production.** [P] CF Fertilisers idled the Billingham ammonia plant in August 2022 and proposed permanent closure in July 2023. CF Industries (Jul 2023)
3. **Ploughing ~57 litres a hectare; direct drilling ~9.** [S] Machinery consultant's figures. 9 ÷ 57 ≈ 16%, so ~84% less fuel (our arithmetic). Farmers Weekly, direct drilling helps slash establishment costs
4. **Total income from farming £8.4bn (2025).** [P/E] Defra, *Agriculture in the UK 2025*, summary. (£1,779m + £1,047m + £925m) = £3,751m ÷ £8,400m ≈ 45%. Income is what is left after all costs, so this compares inputs with farmers' earnings, not with turnover.
5. **Soil degradation £0.9–1.4bn a year.** [P] England and Wales, 2015 study. Graves et al. 2015, *Ecological Economics* 119
6. **14.3% of England's surface waters in good ecological status.** [S] Environment Agency classifications of 6 August 2026, reported by Envirotec (Aug 2026). Farming is one pressure among several, including sewage.
7. **61% of Lough Neagh's phosphorus from farming.** [S] NEW Harmonica model (AFBI and universities), 2021 data; 29% from wastewater. FactCheckNI
8. **Over 40% of Northern Ireland's drinking water; ~£28m treatment, 2022–23.** [S] Al Jazeera (Nov 2023). The £28m is the cost of treating water drawn from the lough, not all of it caused by farming.
9. **88.7% of UK ammonia emissions from farming (2024).** [P] Ammonia reacts in air to form fine particles. National Atmospheric Emissions Inventory, UK agriculture ammonia report 1990–2024
10. **Up to 40% less nitrate leaching.** [S] Deep-rooting cover crops such as fodder radish; guidance, not a single trial; early-drilled covers work best. AHDB, cover crops reduce nutrient losses
11. **Topsoil carbon 0.98% → 1.23% over 28 years (+25%, our arithmetic).** [P] Grass–clover ley rotation on a sandy loam; then a plateau; continuous arable declined. Johnston et al. 2017, Rothamsted
12. **Insect pollination worth ~£630m a year to UK crops.** [P] Defra written answer, 19 September 2025; fruit, vegetables and oilseed rape only, so narrower than older ~£690m estimates. Westminster Brief, Agriculture: pollinators
13. **Savills Virtual Farm.** [M] A modelled 810 ha top-25% East Midlands arable farm, 2024 economics. Overall net margin including SFI, carbon and regenerative premiums: 10% below conventional in year one, 31% above by year six. Food-production margin alone falls 64% in the early years; yields start 26% lower and are still 18% lower by year six. A model of one well-run farm, not a national average. Savills (Jun 2024)
14. **Profit positive after 3 to 5 years.** [M] Six common regenerative practices, crops in scope, European farms; driven by projected yield gains and lower costs, strongest on farms over 55 ha. WBCSD, *Closing the gap* (Mar 2025)
15. **SFI payments: £73/ha no-till; £163/ha summer cover crop.** [P] Sustainable Farming Incentive actions SOH1 and SOH3, 3-year agreements. GOV.UK, SOH1; GOV.UK, SOH3
16. **Urea $400 → $850 a tonne, early 2026 to April.** [P] WTO data blog (10 Jul 2026)
17. **The food reserve one-pager.** Strategic Food Reserve and Food Independence, with its GROW pillar (£310m a year, including £50m to bridge the switch). `pdf/uk-food-reserve-at-a-glance.pdf`

## Who is Planetary Council?

An independent group of citizens working on food security, climate and regenerating the land. We publish our working in the open: every number sourced, every model public, every critique welcome. **Resilience through transparency: tell us where this breaks.**

**Open to** collaboration, consultancy, research and conversation.

**Email** planetarycouncil@protonmail.com

**Open materials**
github.com/planetarycouncil/food-security-climate-regeneration